

Worksheet 19

December 2, 2026

Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

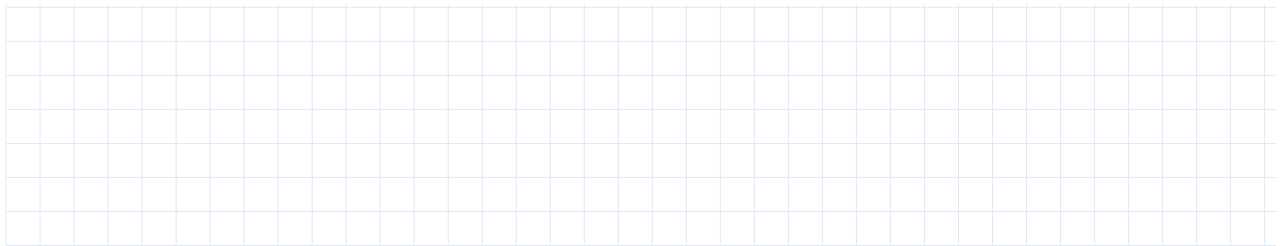
1. Double-angle values

Suppose $\sin \theta = \frac{5}{13}$ and θ is in Quadrant II.

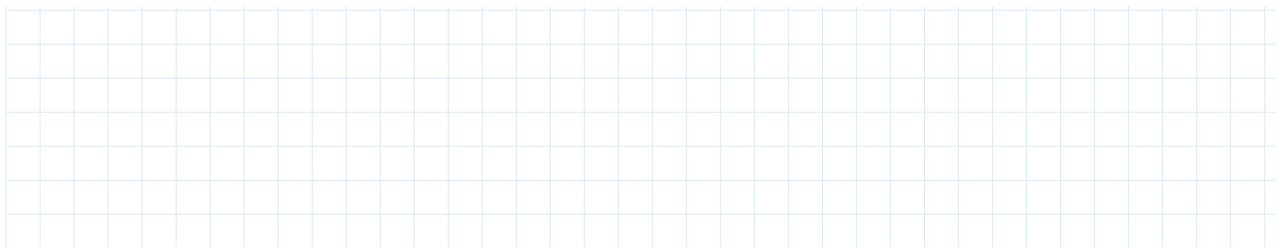
(a) Find $\cos \theta$, including its sign.



(b) Find the exact value of $\sin(2\theta)$.



(c) Find the exact value of $\cos(2\theta)$.



Worksheet 19

December 2, 2026

Name: _____

About 10 minutes total. Begin independently, then discuss. Show your work.

2. Half-angle signs

Suppose $\cos \theta = \frac{3}{5}$ and $\frac{3\pi}{2} < \theta < 2\pi$.

(a) Which quadrant contains $\frac{\theta}{2}$? State the signs of its sine and cosine.

(b) Find exact values of $\sin\left(\frac{\theta}{2}\right)$ and $\cos\left(\frac{\theta}{2}\right)$.